

BASIC MECHANICAL ENGINEERING

MANUFACTURING PROCESSES

Module 4

MANUFACTURING PROCESSES

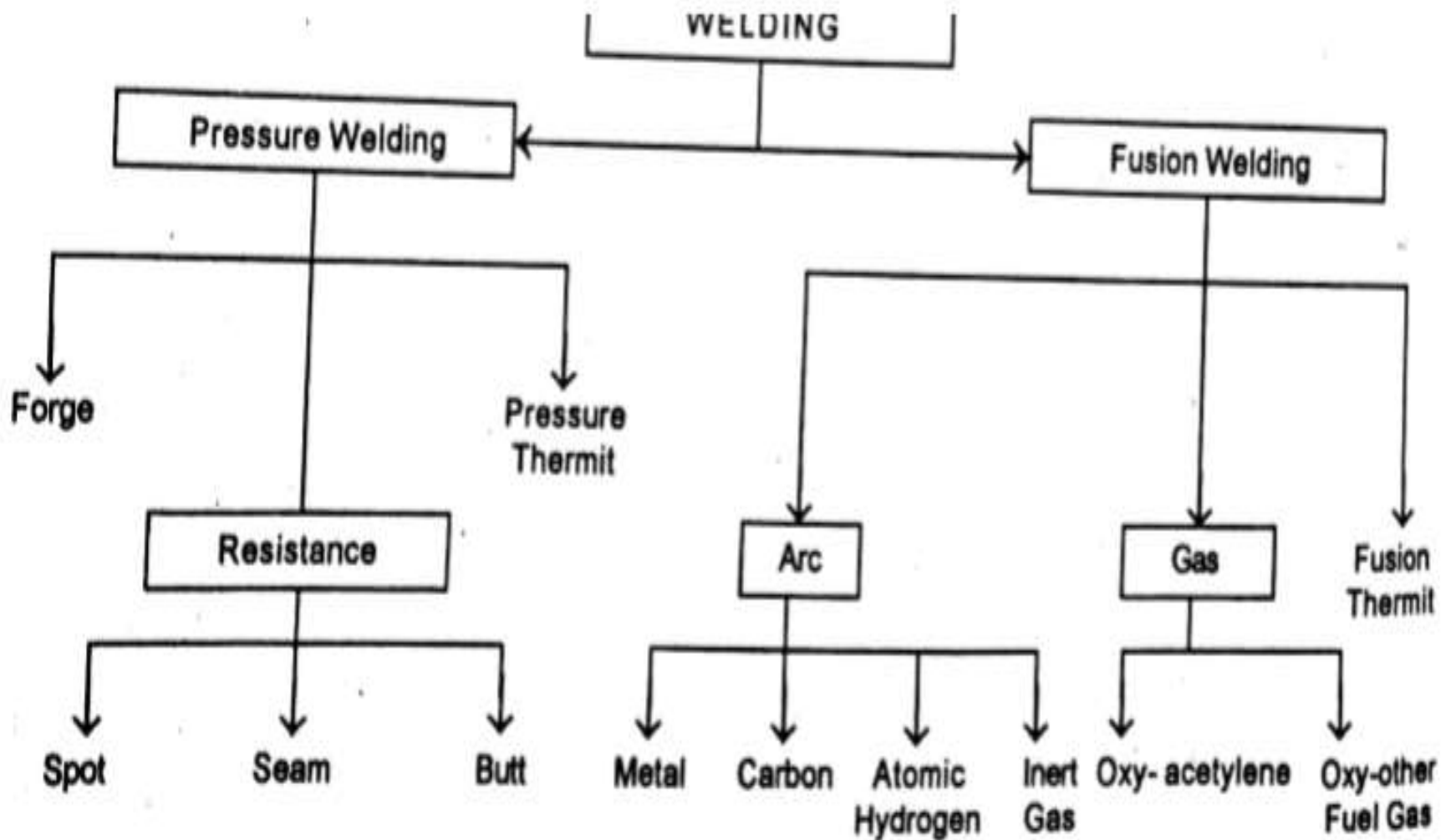
- Manufacturing is the art of transforming raw materials in to finished products.
- It involves step by step procedures by changing shape, size and properties of the material to get a final finished component.
- **Classification**
 1. Shaping Processes
 2. Machining Processes
 3. Joining Processes
 4. Finishing Processes
 5. Property modification process

WELDING

- Is defined as –
- Welding is a process of making a permanent joint by establishing inter atomic bonds between two or more pieces of metal by using heat or heat and pressure.

- Most essential requirement is heat.
- Pressure may or may not be used.
- Filler material may or may not be used.
- Flux is used to resist oxidation and to remove impurities.
- Thus, “Welding is the process of joining similar or dissimilar metals by the application of heat, with or without the application of pressure and addition of filler material”.

Classification Of Welding



Classification Of Welding

- Welding of metals can be divided into two categories.
 1. Plastic Welding (Pressure welding)
 2. Fusion (Diffusion) Welding.
- Plastic Welding -In this type of welding the metals to be joined are to be heated to the plastic state and then forced together by external pressure without the addition of filler material.
Eg. Forge Welding, resistance welding.

- Fusion Welding

In this type of welding no pressure is involved but a very high temperature is produced in or near the joint.

- The metal at the joint is heated to the molten state and a joint is formed as a result of solidification.
- The heat may be generated by electric arc, combustion of gases or chemical action. A filler material may be used during the welding process

Advantages Of Welding

- It produces a permanent joint.
- Overall cost of welding equipment is low.
- Large number of metals can be welded.
- Welding operation can be mechanized.
- Welding operation is economical.
- High corrosion resistance compared to bolting and riveting.

Disadvantages of Welding

- Welding operation distorts workpieces.
- Skilled labour needed
- Weld joints require heat treatment
- Produces chemical and physical changes.
- Gives off harmful radiations.

APPLICATIONS

- Aircraft construction
- Automobile Construction
- Bridges
- Buildings
- Pressure vessels and tanks
- Storage tanks
- Rail Road Equipments
- Piping and Pipelines
- Ships
- Machine tool
- Household
- Heavy machineries

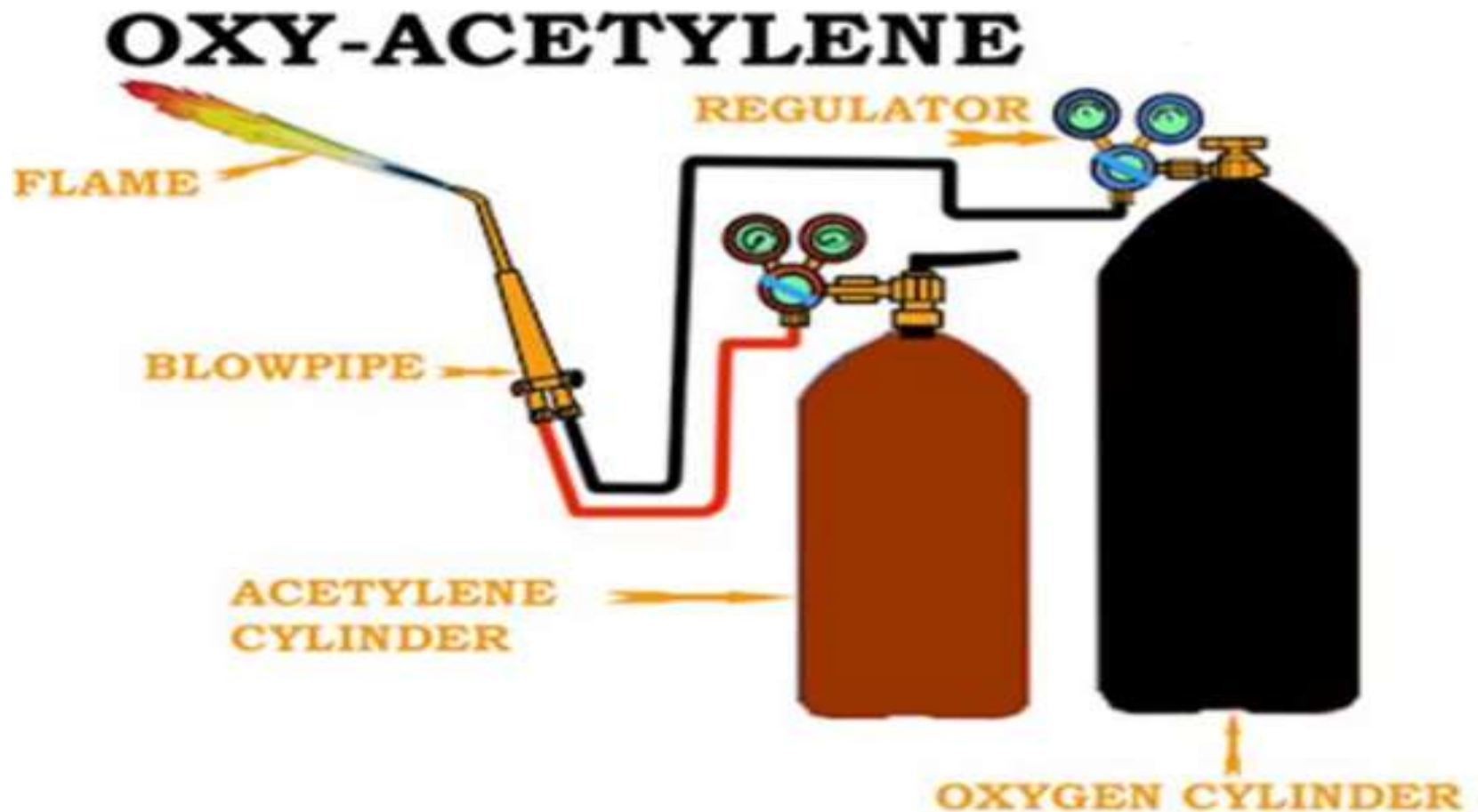
GAS WELDING

- Gas Welding is a Fusion-Welding process.
- It joins metals using the heat of combustion of some fuel gas (acetylene, hydrogen, natural gas) in air or oxygen.
- Temperature produced ranges from 3000 °C to 3400 °C
- The intense heat produced melts and fuses together the edges of parts to be welded, generally with the addition of filler material.

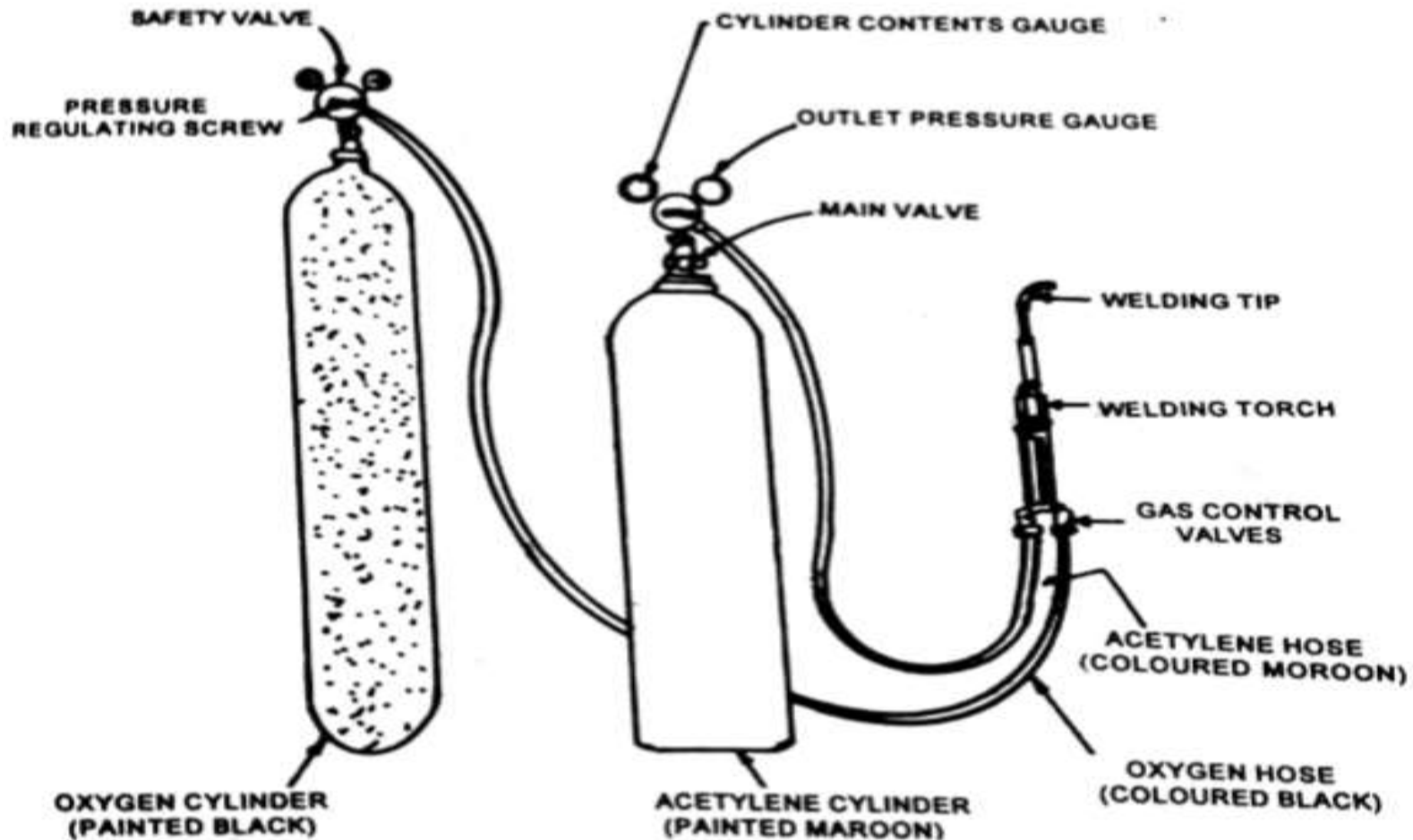
Types of Gas Welding

- Oxy-Acetylene Welding
- Air-Acetylene Welding
- Oxy-Hydrogen Welding

Oxy-acetylene Welding Equipment



Oxy-acetylene Welding Equipment



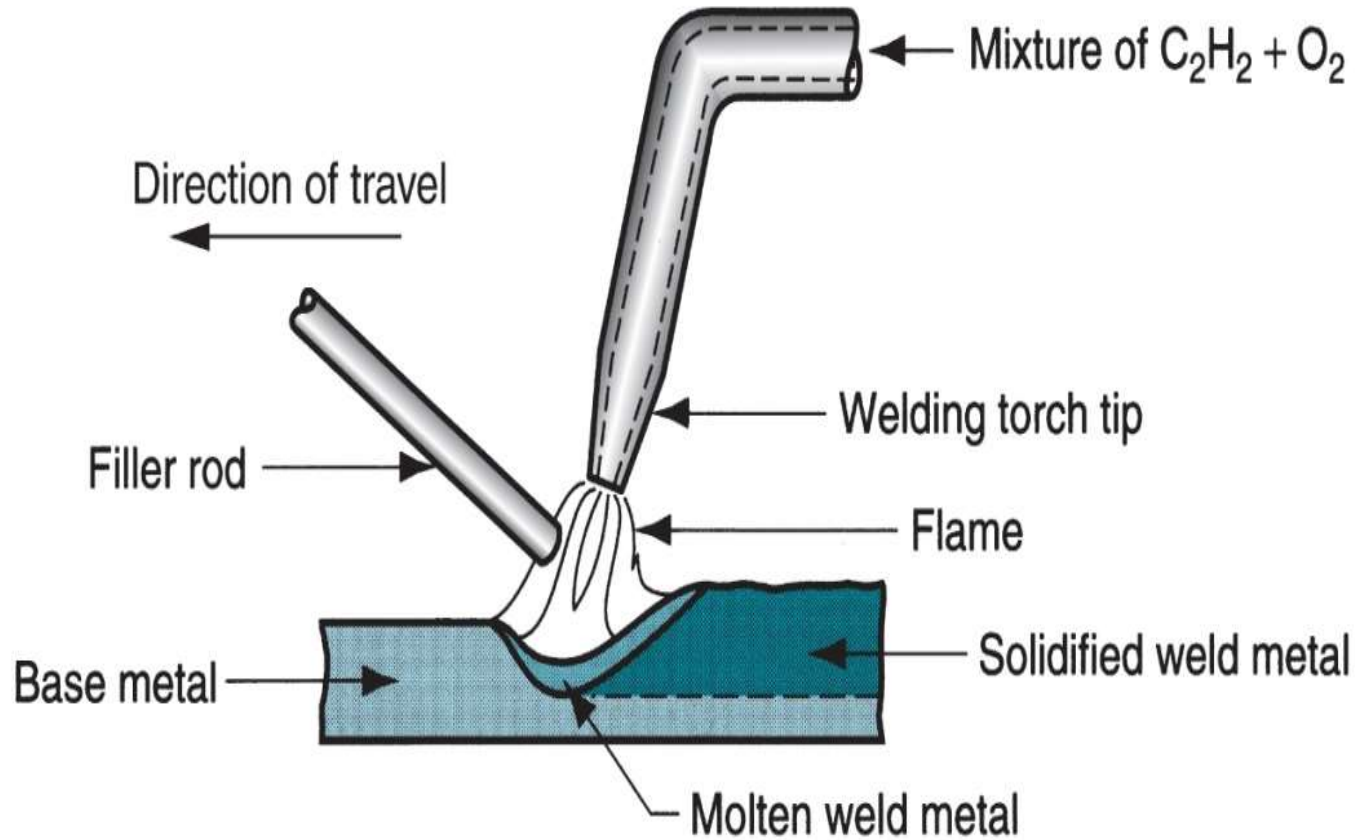
Oxy-acetylene Welding Equipment

- **Acetylene Cylinder** - Used for storing Acetylene
- **Oxygen Cylinder** -Used for storing Oxygen
- **Oxygen and Acetylene pressure Regulators** -The pressure of the gases obtained from the cylinders is considerably higher than the gas pressure required to carry out the welding operation.
- The purpose of regulators is to
 - reduce the pressures of gases
 - to produce steady flow of gases.

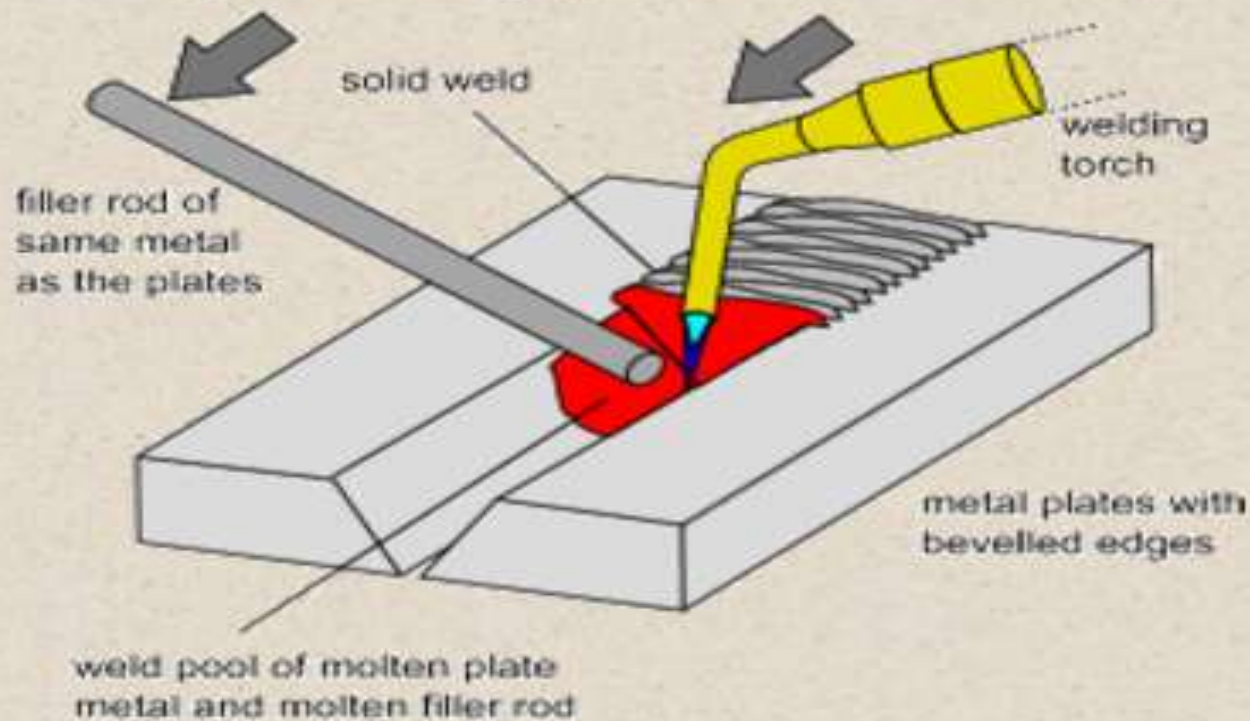
Oxy-acetylene Welding Equipment

- **Welding Rods** - Used as Filler Metal
- **Flux** -Used for removing impurities and oxides.
- **Welding Torch or Blow Pipe** -Oxygen and the fuel gas are mixed in the welding torch. Welding Torch controls the flow of gases to the welding nozzle.

Oxy-Acetylene Welding



Oxy-acetylene welding



Cross section of a completed weld.
The edges of the metal plates
and molten filler rod have fused
together to form the join.

TYPES OF FLAMES

- Neutral flame.

1. All gas is burn
2. gives 100% heat.
3. Ideal flame used in gas welding.

- Reducing flame.

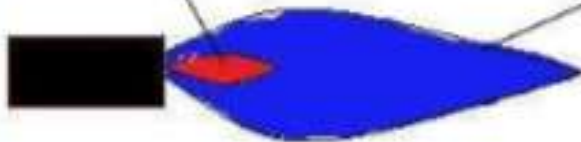
1. It is also called as carburizing flame.
2. It has an excess of acetylene over the oxygen

- Oxidizing flame.

1. It contain more oxygen than any other flame.
2. This is used for oxidizing metal like brass, Zink etc.

Inner Cone (3300 Deg Celcius)

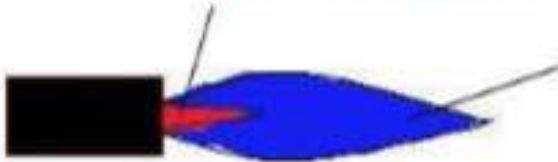
Outer Envelope (1260 celcius)



Neutral Flame

Inner Cone (Pointed)

Outer Envelope (Small and Narrow)



Oxidizing Flame

Bright Luminous
Inner Cone

Acetylene Feather



Carburizing or Reducing
Flame

Blue Envelope

Advantages Of Gas Welding

- Oxy-Fuel gas can be easily controlled.
- Suitable for thin sheets.
- Equipment is portable.
- It can weld most common materials.
- By changing the nozzle the torch can be use for heating, and cutting purposes.
- Cost is low.

Disadvantages of Gas Welding

- Heavy sections cannot be joined.
- Flame temperature is less than that of arc.
- Fluxes used produce fumes that are irritating to eyes, nose and lungs.
- Slower than arc welding Process.
- Distortion to the work piece is more compared to arc welding process.
- Careful handling and storing is required.

ARC WELDING

- Method of fusion welding in which the metals at the joint is heated to molten state by an electric arc
- Arc column is generated between an anode(electrode) and the cathode(metal to be joined)
- Temperature of the arc is about 6000°C to 7000°C

Arc Welding



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graph TD; A((Arc Welding)) --- B((Metal Arc Welding)); A --- C((Carbon Arc Welding)); A --- D((Metal Inert Gas Arc Welding)); A --- E((Tungsten Inert Gas Welding)); A --- F((Plasma Arc Welding)); A --- G((Submerged Arc Welding)); A --- H((Shielded Metal Arc Welding)); A --- I((Electro Slag Welding)); A --- J((Flux Cored Arc Welding));
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**Metal Arc
Welding**

**Carbon Arc
Welding**

**Metal Inert
Gas Arc
Welding**

**Tungsten
Inert Gas
Welding**

**Plasma Arc
Welding**

**Submerged
Arc
Welding**

**Shielded
Metal Arc
Welding**

**Electro Slag
Welding**

**Flux Cored
Arc
Welding**

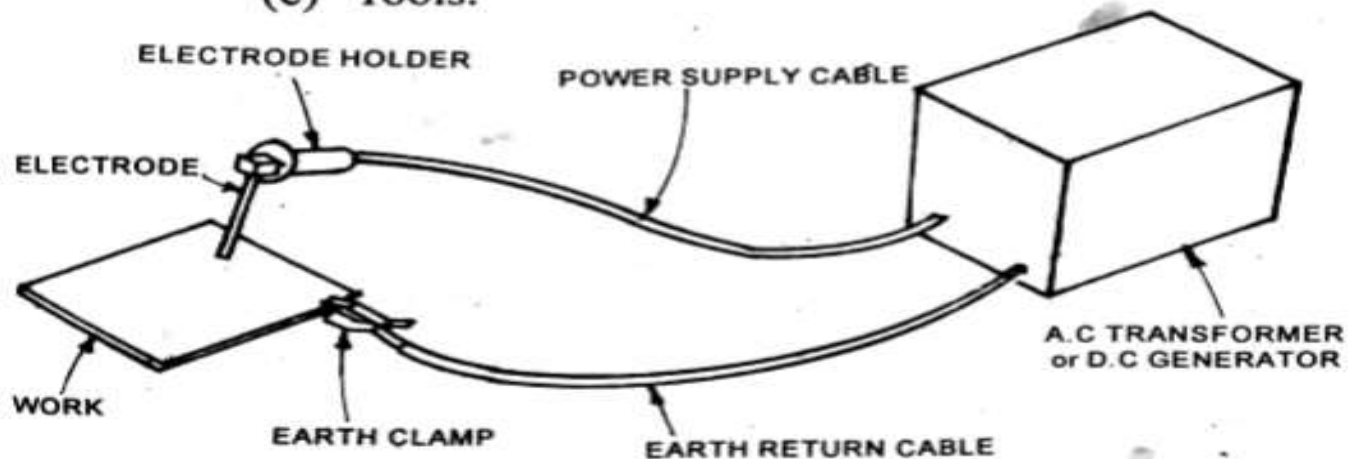
MANUAL METAL ARC WELDING EQUIPMENT

1. Equipment to provide the welding current

- (a) Transformer (for A.C)
- (b) Generator or Rectifier (for D.C)

2. Accessories

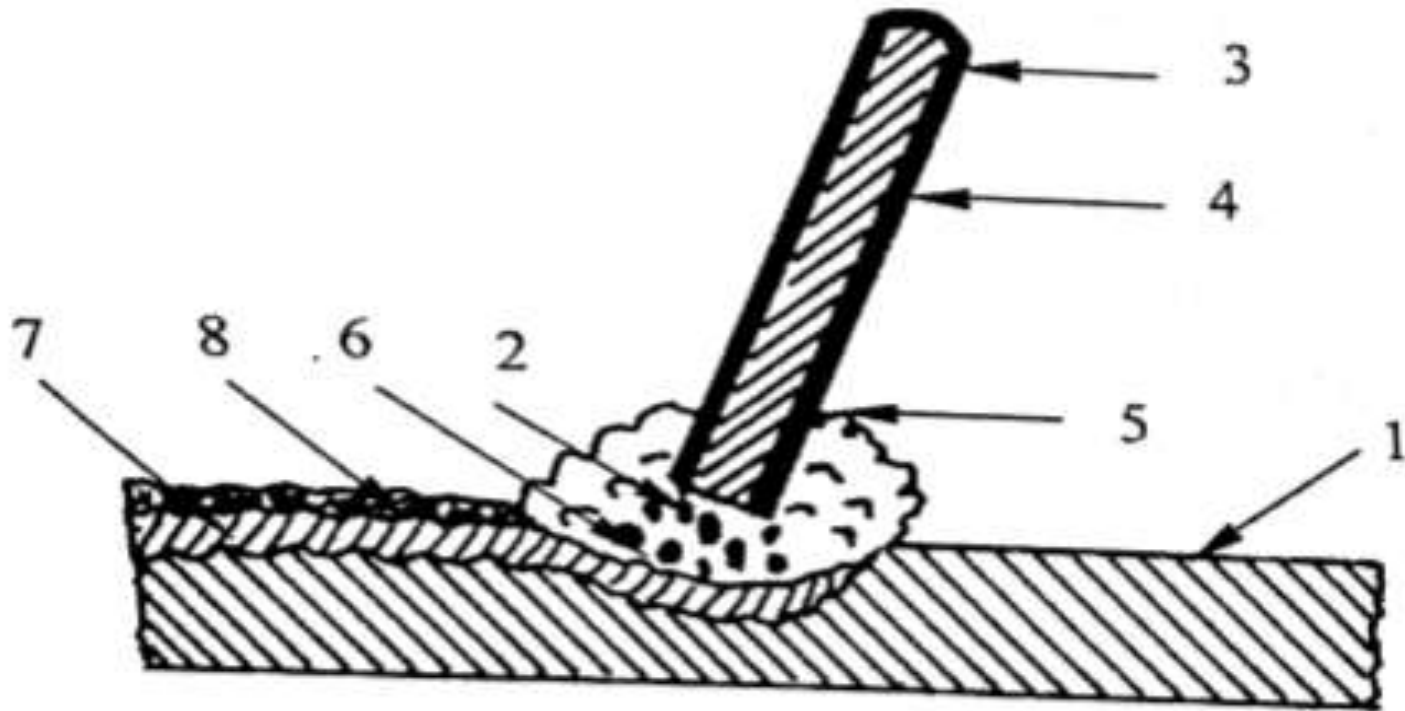
- (a) Electrodes
- (b) Electrode holder
- (c) Cables
- (d) Safety devices and
- (e) Tools.



ELECTRODES

- **Non- consumable electrodes:** carbon, graphite or tungsten(they are not consumed during welding operation)
- **Consumable electrodes:** steel(consumed during welding operation), electrode may be bare(uncoated) or flux coated

ARC WELDING

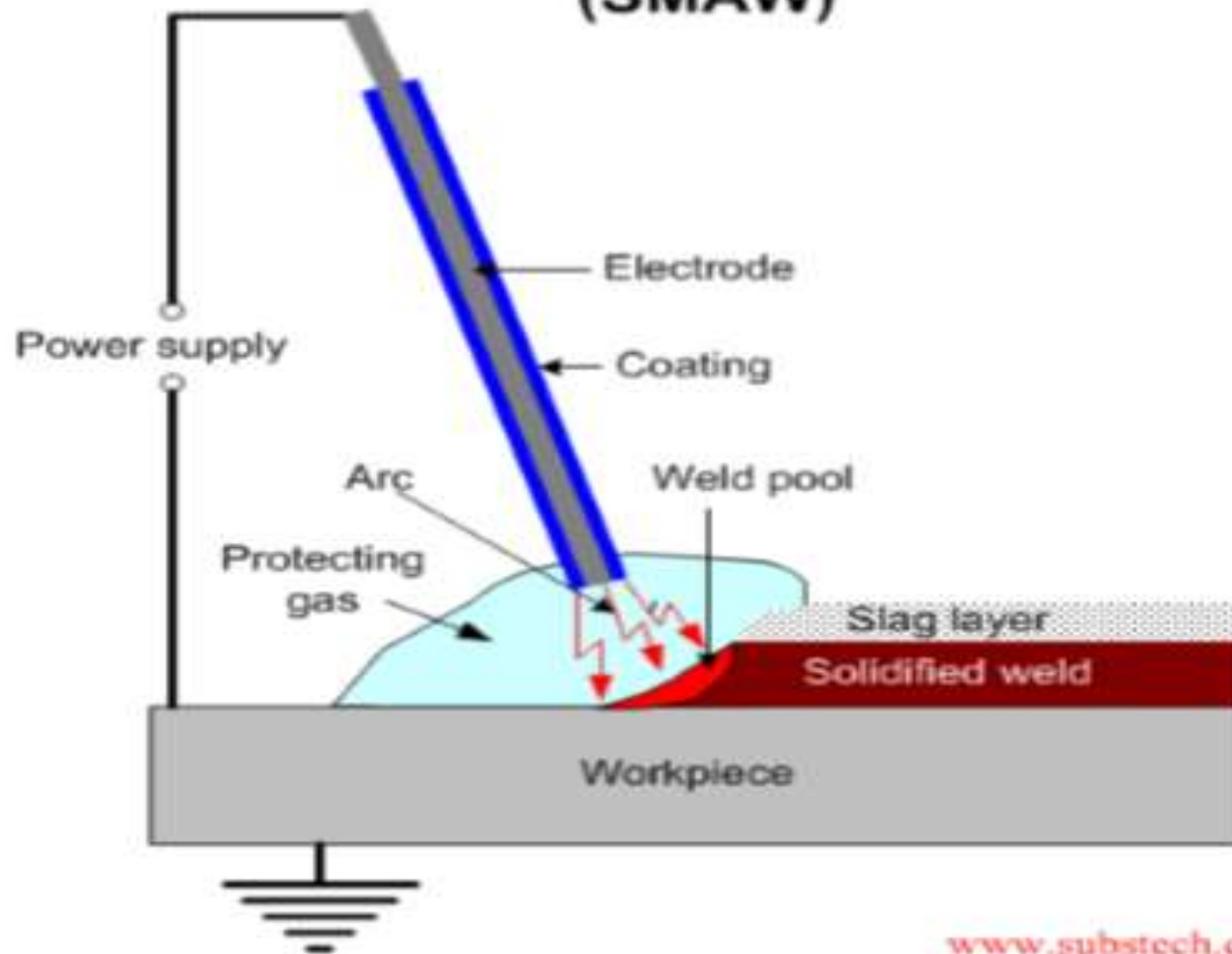


1. Workpiece 2. Arc 3. Electrode 4. Coating
5. Gas shield 6. Molten metal 7. Weld metal 8. Slag

ARC WELDING- MANUAL METAL ARC WELDING

- The heat is generated by an electric arc between base metal and a consumable electrode.
- In this process electrode movement is manually controlled hence it is termed as manual metal arc welding
- In shielded metal arc welding, the protection to the weld pool is provided by covering of
 - a) slag formed over the surface of weld pool/metal and
 - b) inactive gases generated through thermal decomposition of flux/coating materials on the electrode

Shielded Metal Arc Welding (SMAW)



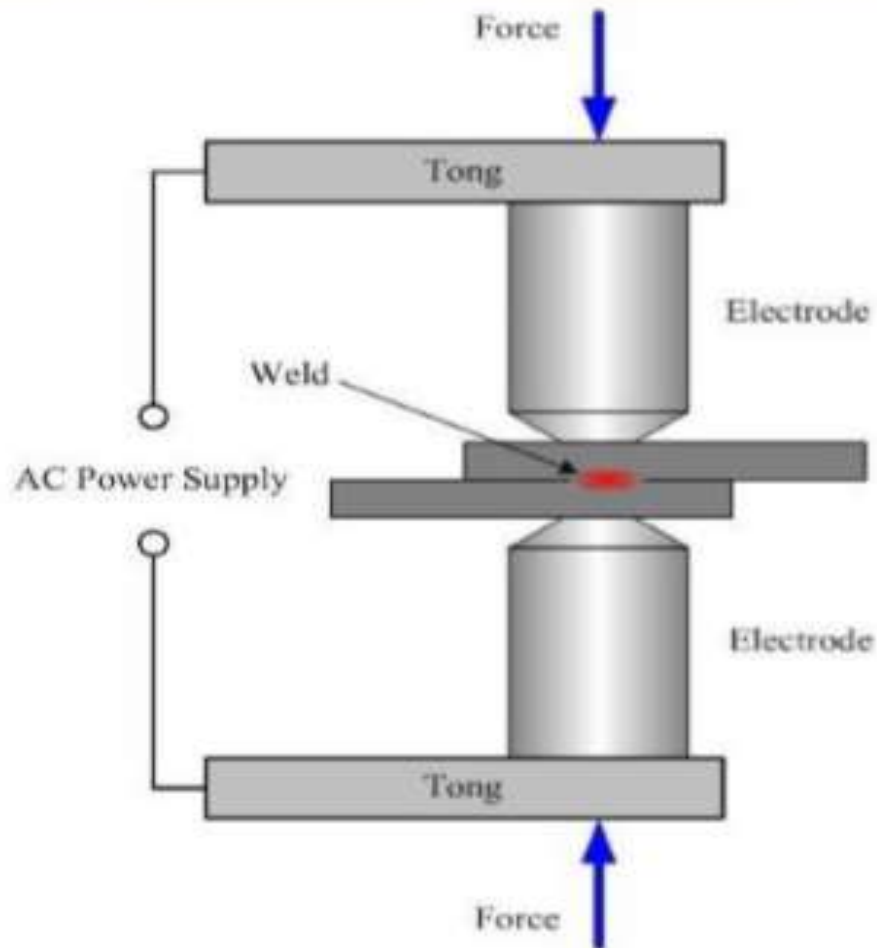
RESISTANCE WELDING

- Parts to be joined are **heated** to a plastic state over a limited area by their resistance to flow of electric current(resistance heating) and then by mechanically pressing together
- **Pressure** is applied continuously till the weld cools down
- It is generally used for joining thin plates and structures

1.RESISTANCE SPOT WELDING

- Pieces are assembled and placed between electrodes and pressure is applied
- When current passes, the pieces are heated at the contact area, and due to the pressure, the contact spot gets welded

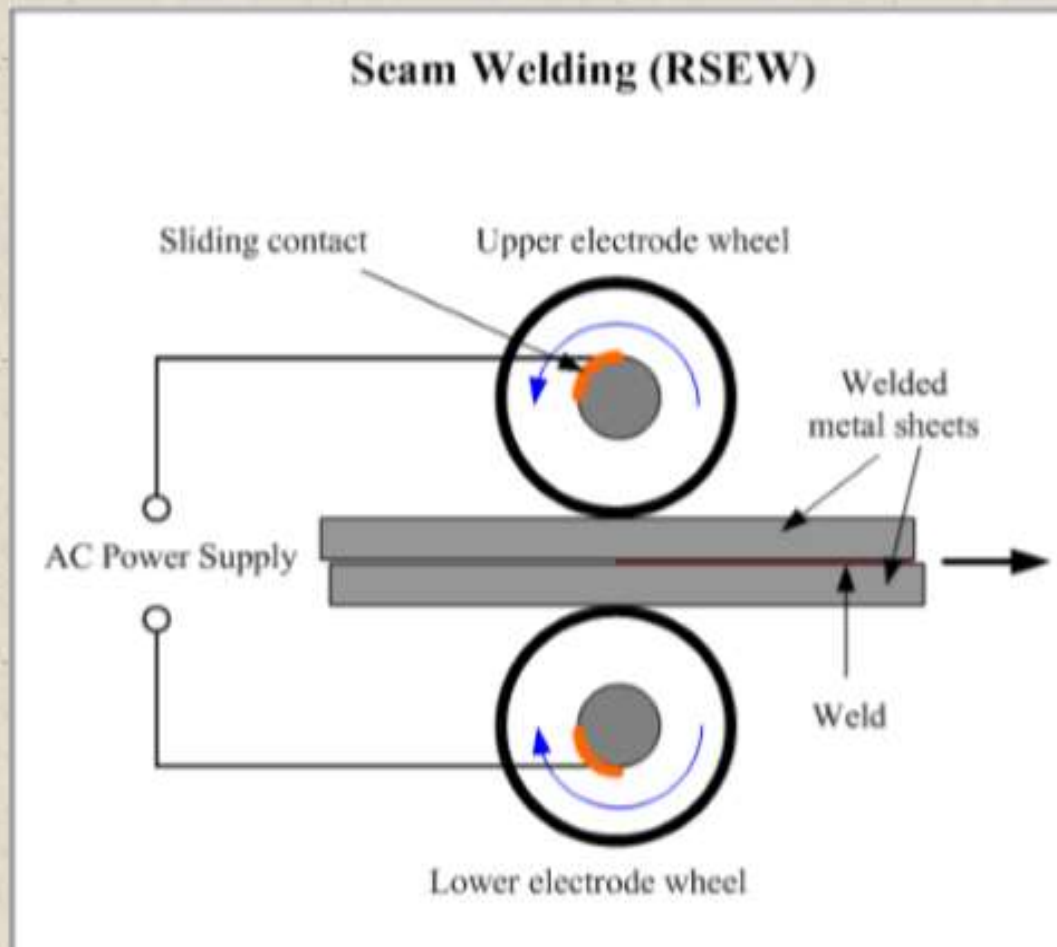
RESISTANCE WELDING- SPOT WELDING



2.RESISTANCE SEAM WELDING

- Similar to spot welding except that the electrodes are disc shaped rollers
- Electrodes roll over the sheet and a continuous weld is obtained
- Timed pulses of current pass through the metal to form the overlapping welds. Welding current is usually higher than the conventional spot welding

RESISTANCE WELDING- SEAM WELDING

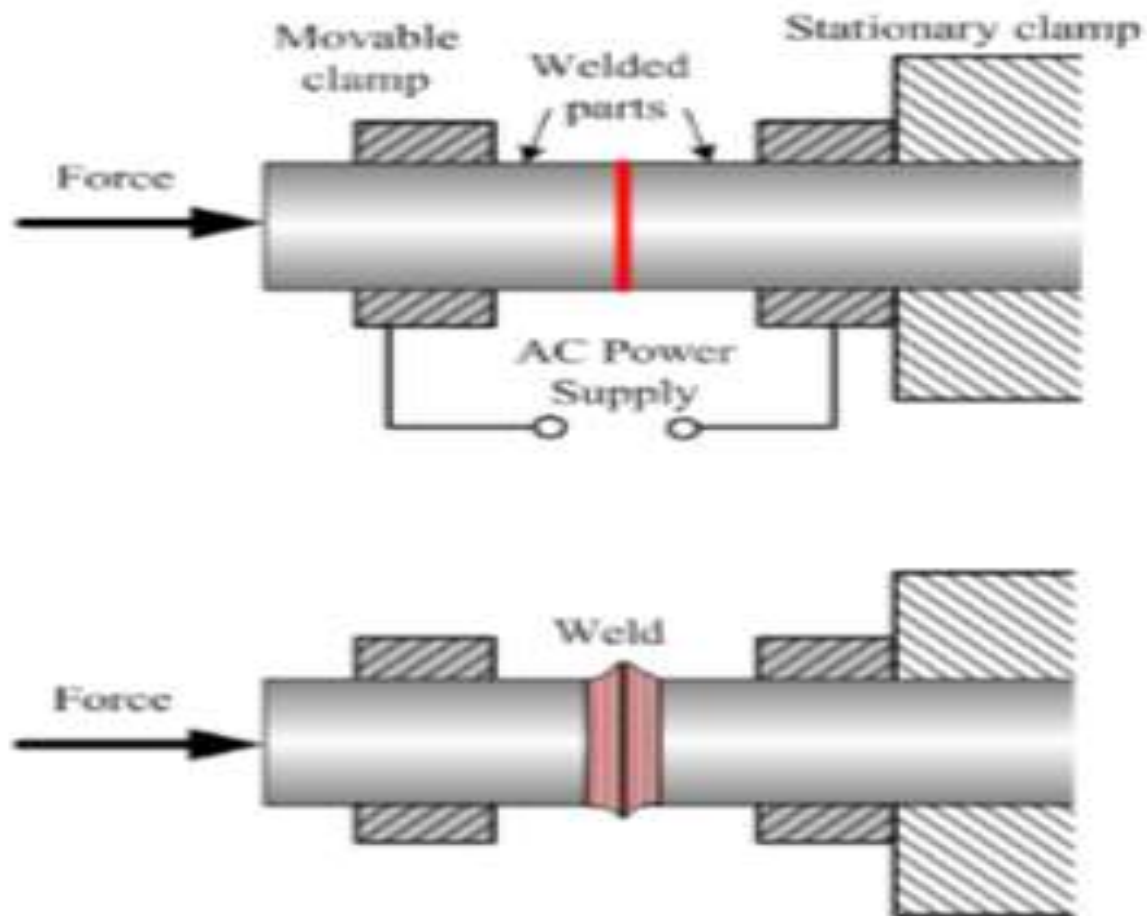


3.RESISTANCE BUTT WELDING

- Heat obtained from the resistance to electric current through the area of contact of two surfaces
- Pressure is applied before heating is started and is maintained throughout the heating period
- Heavy current is passed from one metal piece to other. The resistance to electric current flow heats the faces to fusion temperature
- Both pressure and current are applied throughout the weld cycle

- When the faces of the pieces become plastic, they are pressed together more firmly, upsetting the metal piece to form a dense joint
- Force is released as the welded joint has cooled to the desired temperature
- Here one Clamp is stationary and other one is movable

Butt Welding (UW)



WELDING DEFECTS

- Weld crack
- Porosity
- Undercut
- Spatter
- Slag inclusion
- Poor fusion
- Wrapping and distortion
- Lack of penetration

WELDING DEFECTS

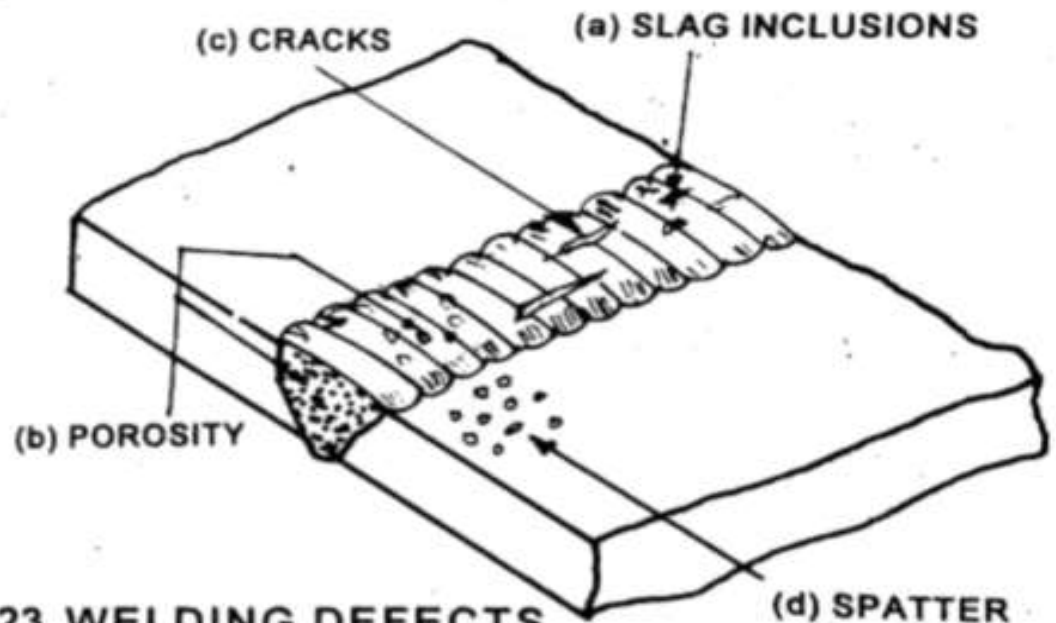
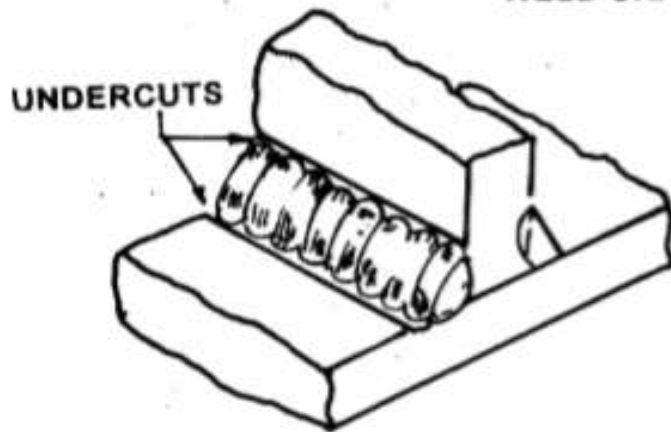
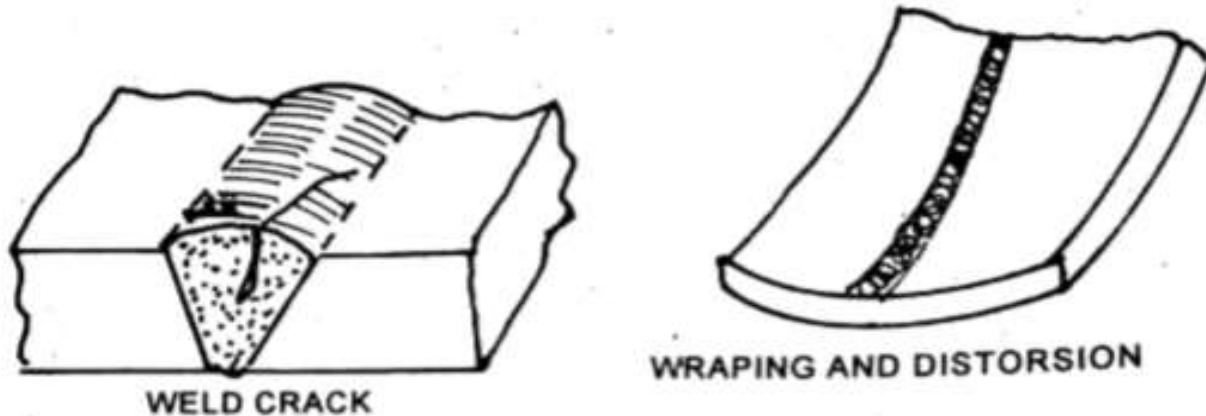
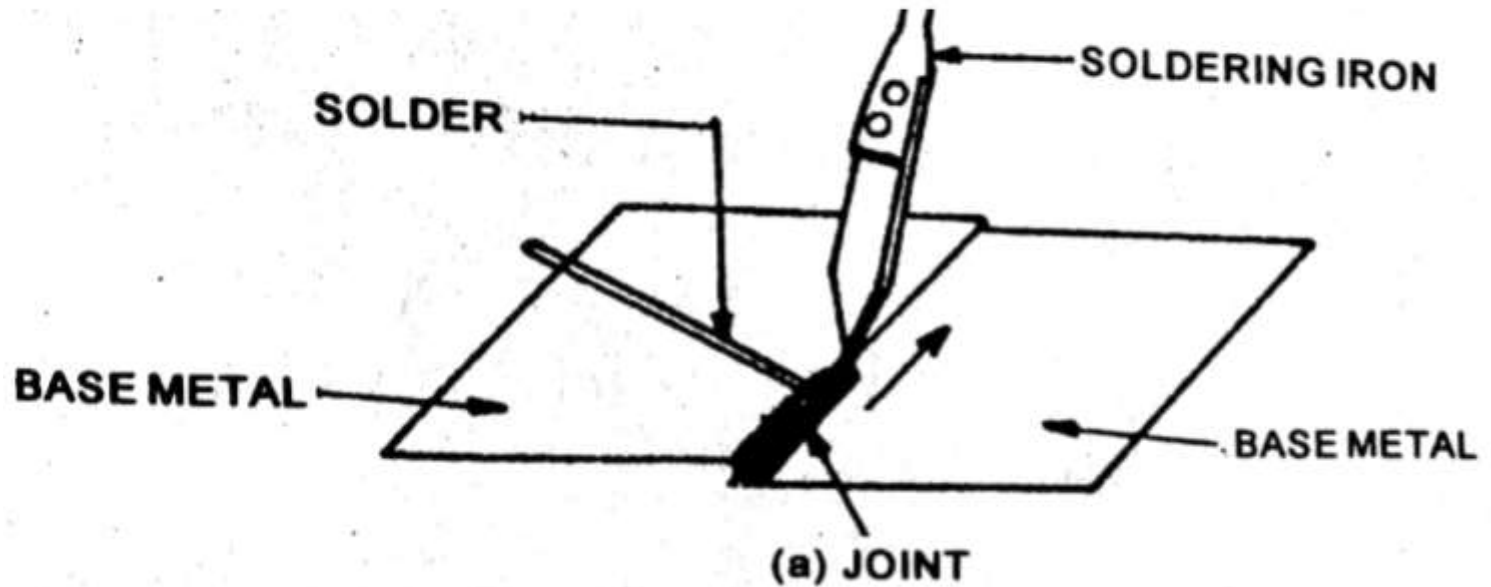


Fig. 4.23 WELDING DEFECTS

SOLDERING

- Method of joining two or more metal pieces by means of a fusible alloy or metal called solder(filler alloy), applied in molten state
- The melting temperature of filler metal is lower than 450°C and also lower than the melting point of the components to be joined

SOLDERING



SOLDERING

- No direct melting of the metals being joined
- During the process, the filler alloy(solder) flows between the two closely adjacent surfaces of the work pieces by capillary action
- Solder is melted by using a soldering iron, which is heated by electrical resistance
- Soldering iron tips are made of copper core plated with iron. The copper is used for heat transfer and the iron plating is used for durability .

TYPES

- **Soft soldering** – generally used in sheet metal work for joining parts which are not subjected to high load or high temperature. The melting point of solder using ranges from 150 to 300 °C
- **Hard soldering** – stronger than soft soldering. Silver alloy with tin is used as hard solder , the melting point of which is 300 to 600 °C

Applications

- Joining wires in electric connections
- Electronic boards and similar small parts which are not subjected to load and temperature

BRAZING

- Metal pieces are joined by heating the closely placed parts and then filler alloy called spelter applied in the molten state which upon solidification produces the desired joint
- Melting temperature of filler metal is more than 450°C but lower than the melting temperature of the components to be joined
- No direct melting of the metals being joined
 - Brazing gives a much stronger joint compared to soldering
- During the process, the filler alloy(spelter) flows between the two closely adjacent surfaces of the work pieces by capillary action

TYPES

- TORCH BRAZING
- FURNACE BRAZING
- RESISTANCE BRAZING
- INDUCTIVE BRAZING

APPLICATIONS - Fastening of pipe fittings, stove burners, carbide tip on tools, radiators, heat exchangers, electrical parts

BRAZING

